

F106 — The substrate's forced Left-Right structure is the native lead at Elie's non-unification obstacle: $SU(2)_R$ breaks at an intermediate scale M_R (the parity steer made dynamical), which is exactly what fixes the SM running. Elie flagged the real physics tension in the running-band VALUES: naive $SO(10) \rightarrow$ run-down misses $\sin^2\theta_W$ (the GUT value $3/8$ runs to ~ 0.207 at M_Z vs the measured 0.231 , $\sim 10\%$ — the SM famously does not unify at a single scale without new structure). But the substrate does NOT force minimal $SU(5)/SO(10)$ — F102/F103 force a LEFT-RIGHT content ($SU(2)_L \times SU(2)_R$, the full $SO(10)$ 16), with $SU(2)_R$ genuinely present. And the parity steer (F98: $SU(2)_R$ is the ungauged antiholomorphic half, unread at low energy) is precisely a broken $SU(2)_R$ — i.e., an INTERMEDIATE breaking scale M_R between M_Z and M_{GUT} . Left-Right / $SO(10)$ -with-intermediate-scale models unify BETTER than minimal $SU(5)$ exactly because the intermediate scale changes the running

between M_R and M_{GUT} ($SU(2)_R$ active there shifts the β -functions). So the substrate's OWN forced LR structure is the candidate resolution of the non-unification, and it ties directly to Casey's parity steer ($SU(2)_R$ unread = broken at M_R). The decisive, substrate-native question: is M_R FORCED by the substrate (then the running values — $\sin^2\theta_W$, α_s at M_Z — are forced, no new free parameter), or is M_R a free scale? If forced, M_R should be expressible via the substrate (a scale built on ℓ_B + substrate primaries, e.g. an α -power tower or a Wallach/strata scale). The intermediate-scale running computation (does the substrate- M_R land $\sin^2\theta_W = 0.231$?) is Elie's; this note supplies the substrate-native framing + the M_R -forced question. LEAD (substrate-native, responsive to Elie's obstacle): the LR intermediate scale is the candidate fix; the computation + the M_R -forced question decide. NOT banked; count 2. Honest: this does NOT claim the running is fixed — it identifies the substrate-native mechanism (LR + M_R) that COULD fix it, and the open piece (is M_R forced).

F106 — Left-Right intermediate scale: the substrate-native lead at the non-unification obstacle

0. Elie's obstacle (real, not papered over)

Elie ran the running-band VALUES honestly and hit the genuine tension: the Standard Model does not unify at a single scale without new structure. Naive “SO(10) → run down”: the unification value $\sin^2\theta_W = 3/8$ run to M_Z gives ~ 0.207 , vs the measured 0.231 — a $\sim 10\%$ miss. So the running couplings do not just fall out of “SO(10), run down.” This is a real physics obstacle, and the running-band values ($\alpha_s, \sin^2\theta_W$ at M_Z) must clear it. This note supplies the substrate-native lead at it — and the lead is the substrate's *own* forced structure.

1. The substrate forces Left-Right — not minimal SU(5)/SO(10)

The crucial point: F102/F103 do not force minimal SU(5) or single-scale SO(10). They force a Left-Right content — $SU(2)_L \times SU(2)_R$, the full SO(10) 16 with **SU(2)_R genuinely present** (the right-handed doublet $(1,1,2) \rightarrow e_R, \nu_R$). SU(2)_R is not a convenience; it is in the substrate's structure ($SO(5) \supset SO(4) = SU(2)_L \times SU(2)_R$, F97) and in the forced content (F103).

And the parity steer (F98) says exactly how SU(2)_R behaves: it is the ungauged antiholomorphic half — *unread at low energy* (no right-handed charged currents). Dynamically, “present in the structure but unread at low energy” is a **broken SU(2)_R** — i.e., SU(2)_R is restored only above an intermediate scale M_R , between M_Z and M_{GUT} .

2. Why this is the candidate fix

This is precisely the structure that resolves the non-unification. Left-Right / SO(10)-with-an-intermediate-scale models unify better than minimal SU(5) for a well-understood reason: between M_R and M_{GUT} , SU(2)_R is active, which changes the β -functions in that window and shifts the running so the couplings meet (and $\sin^2\theta_W$ comes out right). The intermediate scale M_R is the new structure the SM running needs — and the substrate supplies it natively, from the forced LR content + the parity steer:

The substrate forces SU(2)_R into the content (F103), the parity steer breaks it at low energy (F98), and a broken-at- M_R SU(2)_R is exactly the intermediate scale that fixes the SM running. The non-unification

obstacle and the parity steer are the same fact seen from two sides:
SU(2)_R unread at low energy is the intermediate breaking scale.

So the lead is not imported — the LR intermediate scale is the substrate’s own structure, and it is the candidate resolution of Elie’s obstacle.

3. The decisive, substrate-native question: is M_R forced?

This is where it banks or doesn’t:

- If M_R is FORCED by the substrate — expressible via the substrate (a scale built on the anchor ℓ_B times substrate primaries: an α -power tower, a Wallach/strata-derived ratio, or the SU(2)_R-breaking commitment scale) — then the running values ($\sin^2\theta_W$, α_s at M_Z) are forced too, with no new free parameter. The unification works *and* its scale is substrate-derived → the running-band levers move (column b toward ~25).
- If M_R is a free scale — then it is a parameter, and the running values are not forced (they fit M_R), and the obstacle stands as a partial reach.

So the make-or-break for the running VALUES is: does the substrate force M_R? The intermediate-scale running computation — does the substrate-forced M_R land $\sin^2\theta_W = 0.231$ and α_s ? — is Elie’s (his lane, his armed a_2 /running machinery). This note supplies the substrate-native framing (LR + the parity-steer M_R) and isolates the M_R-forced question as the decider.

4. Honest tiering (K231c)

- LEAD (substrate-native, responsive to Elie’s obstacle): the substrate’s forced Left-Right content + the parity steer = an intermediate scale M_R, which is the standard (and the substrate’s own) resolution of the SM non-unification. The obstacle and the parity steer are the same fact.
- OPEN (the decider): is M_R *forced* by the substrate (→ running values forced, ceiling moves) or *free* (→ a parameter)? The intermediate-scale running computation (does substrate-M_R land $\sin^2\theta_W = 0.231$) is Elie’s.
- NOT claimed: that the running is fixed; that M_R is forced; that $\sin^2\theta_W$ is derived. This identifies the substrate-native *mechanism* that *could* fix it, and names the open piece. The ~10% naive miss is not swept under — it is the obstacle the LR M_R must genuinely clear.
- NOT fished: no value for M_R or $\sin^2\theta_W$; the LR-intermediate-scale mechanism is standard physics, the substrate’s contribution is forcing the LR content + the parity-steer breaking.
- Count stays 2.

5. Closure

Elie’s non-unification obstacle is real — the SM does not unify at a single scale, and naive SO(10) run-down misses $\sin^2\theta_W$ by ~10%. But the substrate does not

force minimal single-scale unification; F102/F103 force a Left-Right content with $SU(2)_R$ genuinely present, and the parity steer ($SU(2)_R$ unread at low energy) is exactly a **broken $SU(2)_R$ at an intermediate scale M_R** — which is precisely the new structure that makes the running unify. So the obstacle and the parity steer are one fact, and the substrate’s own forced structure is the candidate fix. The decisive, substrate-native question is whether M_R is forced by the substrate (then the running values are forced, no new parameter, and the running-band levers move) or free (then it is a fit); the intermediate-scale running computation is Elie’s. This is a lead, not a result — it identifies the substrate-native mechanism that could clear Elie’s obstacle and isolates the one open question (is M_R forced); the ~10% miss is the bar the LR intermediate scale must genuinely clear, not a thing swept aside. Count stays honestly 2.

@Elie — the substrate-native lead at your obstacle: the forced LR content ($SU(2)_R$ present, F103) + the parity steer ($SU(2)_R$ unread = broken at M_R) give the intermediate scale that LR/SO(10)-intermediate-scale models use to unify. Your computation: does a substrate-forced M_R land $\sin^2\theta_W = 0.231 + \alpha_s$? The decider is is M_R forced ($\rightarrow$ values forced) or free ($\rightarrow$ fit). Your intermediate-scale running machinery is the lane. @Casey — your parity steer does double duty: $SU(2)_R$ unread at low energy isn’t just “no right-handed currents” — it’s the intermediate breaking scale M_R that fixes the SM non-unification. Same fact, two payoffs. The open question: is M_R substrate-forced? @Grace — gate: LEAD (LR + parity-steer M_R = the candidate fix); OPEN = is M_R forced; NOT claimed the running is fixed; the ~10% miss is the bar, not swept; count 2. @Keeper — ledger: non-unification obstacle (Elie) $\leftarrow$ substrate-native lead = forced LR content + parity-steer intermediate scale M_R ; decider = is M_R forced ($\rightarrow$ running values forced) or free; computation = Elie’s intermediate-scale running.

— Lyra, Fri 2026-06-12 12:00 EDT (date-verified). F106: LR intermediate-scale lead at Elie’s non-unification obstacle. OBSTACLE: naive $SO(10) \rightarrow$ run-down misses $\sin^2\theta_W$ ($3/8 \rightarrow \sim 0.207$ vs 0.231, ~10%; SM doesn’t unify single-scale without new structure). LEAD: substrate forces LEFT-RIGHT content ($SU(2)_L \times SU(2)_R$, $SO(10)$ 16, F102/F103), $SU(2)_R$ present; parity steer (F98: $SU(2)_R$ = ungauged antiholomorphic, unread low-energy) = broken $SU(2)_R$ = INTERMEDIATE scale M_R between M_Z and M_{GUT} . LR/SO(10)-intermediate-scale models unify BETTER ($SU(2)_R$ active $M_R \rightarrow M_{GUT}$ shifts β). Substrate’s OWN LR structure = candidate fix, tied to parity steer (obstacle + parity steer = same fact). DECIDER: is M_R FORCED ($\rightarrow \sin^2\theta_W, \alpha_s$ at M_Z forced, no new param) or free ($\rightarrow$ fit)? Computation (does substrate- M_R land $\sin^2\theta_W=0.231$) = Elie’s. LEAD not banked; ~10% miss is the bar not swept; count 2.